Recent Progress In Quantum Algorithms

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AQIS07 [Kyoto, Japan, September

The Goal of This Talk



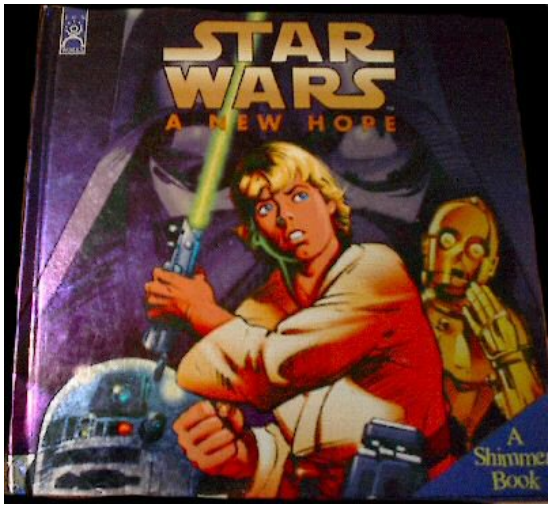
80s-early 90s: foundational work

1994: Shor's algorithm

1996: Grover's algorithm

Last decade: lots of very very smart people worked on coming up with new quantum algorithms which offered exponential speed-ups, with no extraordinary breakthroughs

Today I hope to show you that all hope is not lost, that progress (not a breakthrough!) can be made, even by slow witted physicists like myself.



Jet Lag = Quantum Algorithms Nightmares

Tlon, Uqbar, Orbis Tetris Times, April 1, 20

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HELP WANTED Quantum programmer. Only skill necessary is ability to type in large RSA keys. Salary commiserates with experience.

HELP WANTED Real estate agent. Best market since bubble popped in 07. Join a dedicated

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But what to work on?



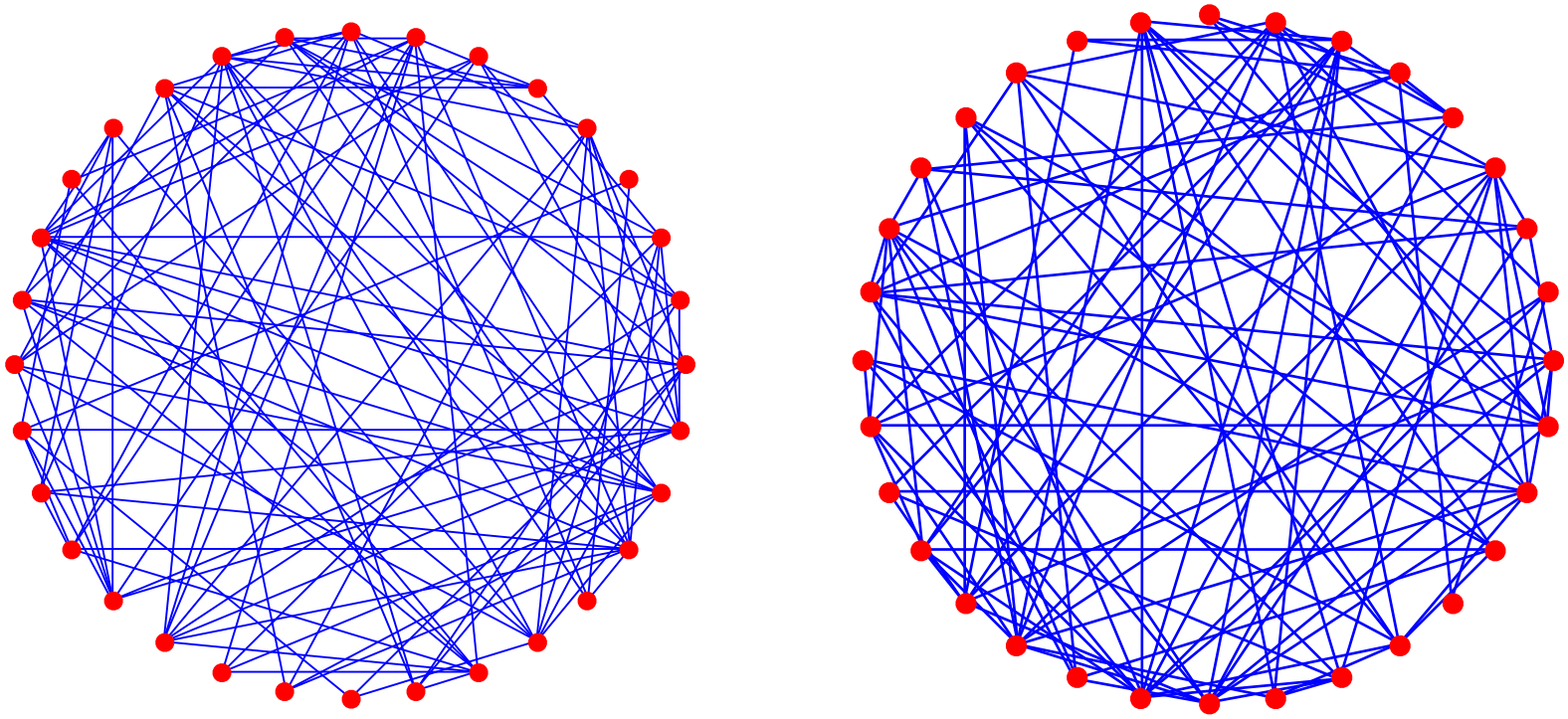
Grover speedups? To easy?

Just right algorithms in BQP?

NP-complete? To hard

Graph Isomorphism

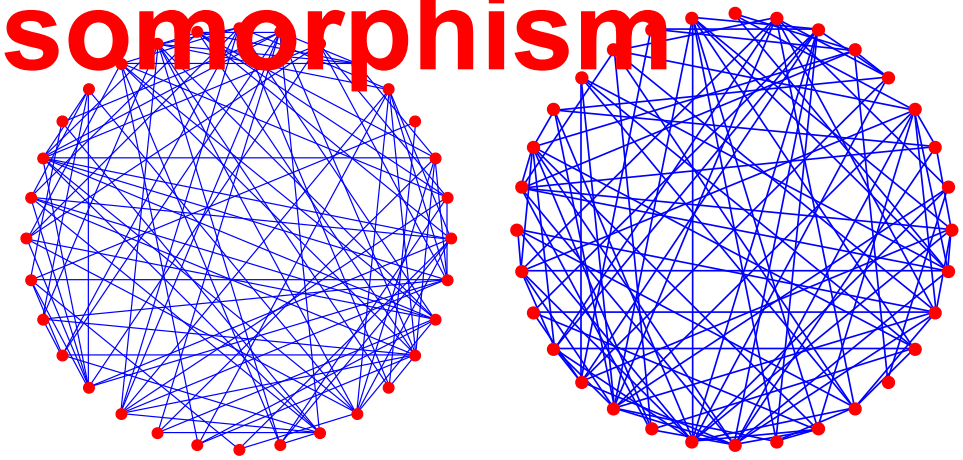
Are these two graphs isomorphic?



Graph isomorphism is in NP∩coAM

- No known polynomial time algorithm
- Probably not NP-complete (⇒collapse polynomial hierarchy)
- Factoring in related class NP∩coNP

Graph Isomorphism



$f(\pi) =$ Adjacency matrix of *G1*∪*G2* with edges permuted by π

[For simplicity, assume graphs rigid]

If isomorphic, exists a subgroup {e,π} which leaves function constant
on all left cosets cosets of this subgroup {g,gπ}.

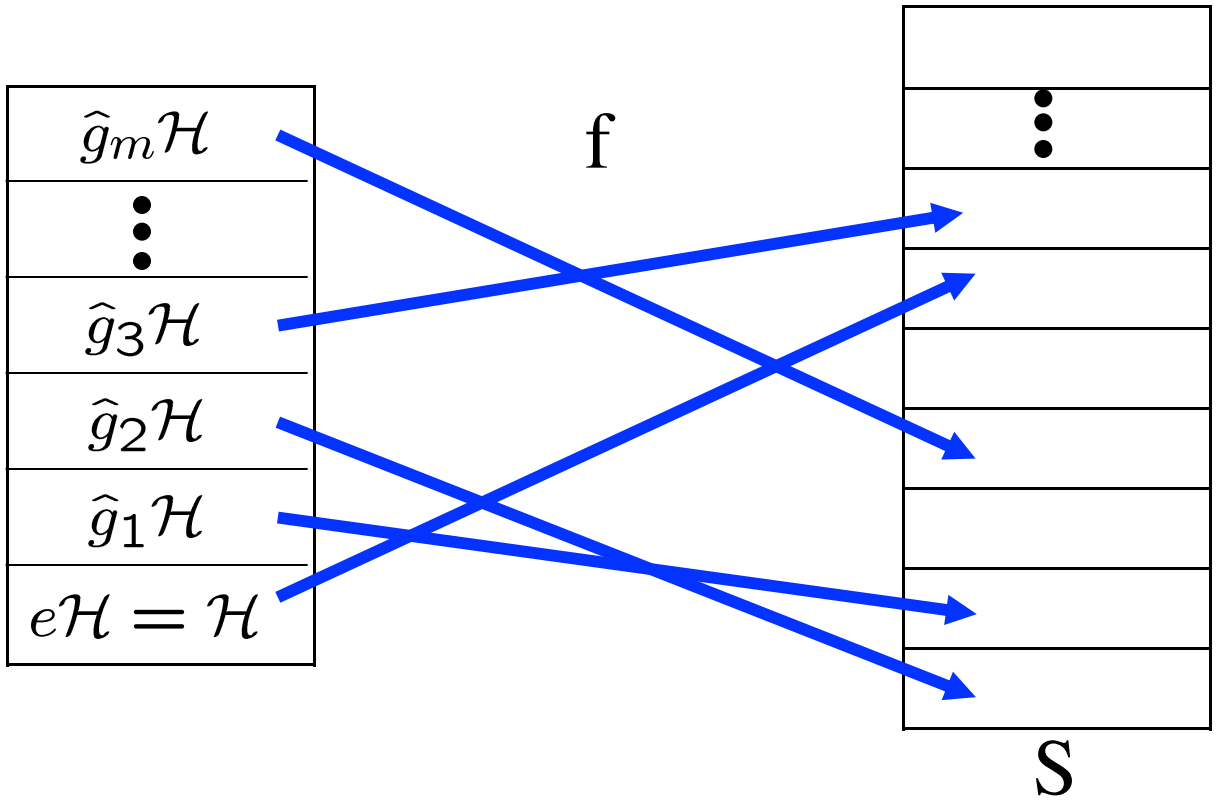
This is an instance of the hidden subgroup problem.

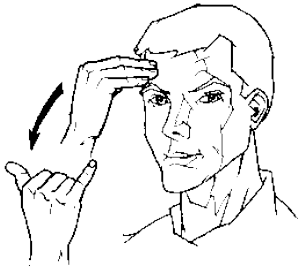
The Hidden Subgroup Problem

Setup: Fix a group $\mathcal{G}$(known) and a subgroup $\mathcal{H} \subset \mathcal{G}$(unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$ We say that f hides $\mathcal{H}$

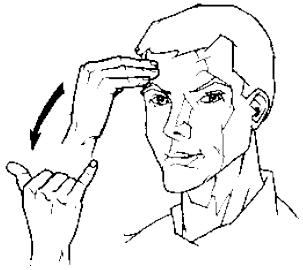
Problem: Find a generating set for $\mathcal{H}$ ("identify $\mathcal{H}$").

Efficient: If runs in time polylog($|\mathcal{G}|$)





Why the Fuss?



Solved Problems (Shor 1994)

efficient algorithm for HSP over $\mathbb{Z}_N$ → efficient algorithm for factoring whole numbers

efficient algorithm for HSP over $\mathbb{Z}_N \times \mathbb{Z}_N$ → efficient algorithm for the discrete logarithm problem

Open Problems

efficient algorithm for HSP over $\mathcal{S}_n \wr \mathbb{Z}_2$ (or $\mathcal{S}_{2n}$) → efficient algorithm for the graph isomorphism problem

efficient "standard" quantum algorithm for HSP over dihedral group $\mathcal{D}_N$ → efficient algorithm for the poly(*n*) unique shortest vector in a lattice problem

A Strawman

BQP, PP, NP, PostBQP, MIT….



Scott Aaronson of MIT

What makes you think the hidden subgroup approach to solving graph isomorphism is the correct approach to fining a quantum algorithm for this problem?

Philosonomico

What is the difference between classical and quantum computing?

[The fair comparison is with probabilistic classical computing]

Configurations:	Σ^n $01001\cdots01$	Σ^n $\|01001\cdots01\rangle$
State Descriptions	$p \in (\mathbb{R}^{\|\Sigma\|})^n$	$\psi \in (\mathbb{C}^{\|\Sigma\|})^n$
Gates:	stochastic matrices $p_{out} = M p_{in}$ $M_{i,j} \in \mathbb{R}$ $M_{i,j} \geq 0$ $\sum_i M_{i,j} = 1$	unitary matrices $\psi_{out} = U \psi_{in}$ $U_{i,j} \in \mathbb{R}$ $UU^\dagger = I$

Some Outrageous Statements

Classical Reversible Deterministic Computation:

Is nothing more than the study of the properties of certain generating sets of a very large permutation group

Classical Reversible Probabilistic Computation:

Leads to doubly stochastic matrices…and what?

Quantum Computation:

Leads to unitary matrices

We can now apply all the tools and trades of ***representation theory of groups*** to the study of computation

Group representation theory is the natural language of the most natural model of computation

Group Representation Theory

Group: set of elements with binary multiplication operation (satisfying closure, associativity, identity and inverse)

(Unitary) representation of a group

$$D : \mathcal{G} \rightarrow \mathsf{U}(d)$$

which is a homomorphism

$$D(g_1)D(g_2) = D(g_1 g_2)$$

For finite and compact groups we can decompose representation into irreducible representations (irreps)

$$D(g) = \bigoplus_{\mu} I_{n_\mu} \otimes D_\mu(g)$$

Irreps have no invariant subspaces except for the empty and full subspace. They are the “primitives” of group representation theory.

Comparison of Symmetry

$T := 0 \to 1, 1 \to 0$

Classically, suppose we have a probability distribution with

$$Tp = p$$

What can you do?

Quantum mechanically, suppose we have a density matrix with

$$T\rho T^\dagger = \rho$$

Then there is “more” information about the state in a new basis

$$\rho = p|+\rangle\langle+| + (1-p)|-\rangle\langle-|$$

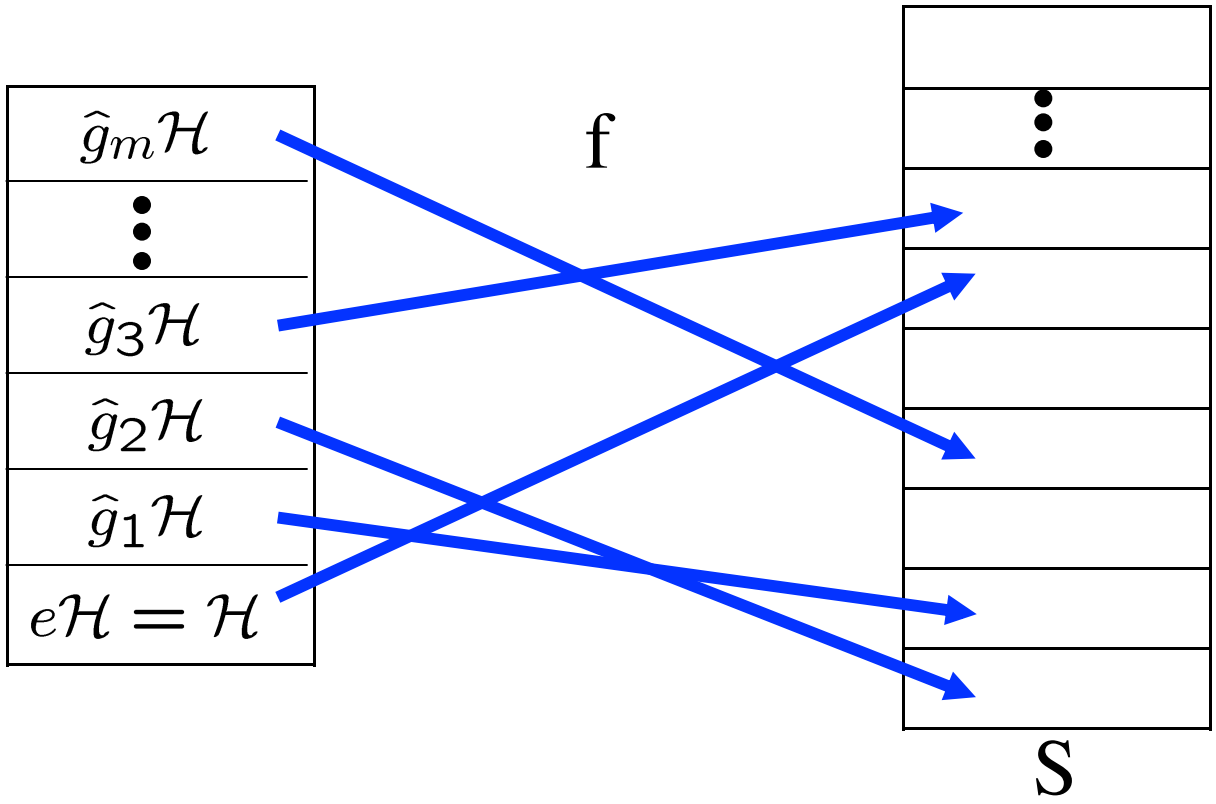
$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$$

The Hidden Subgroup Problem

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Problem: Find a generating set for $\mathcal{H}$ ("identify $\mathcal{H}$").

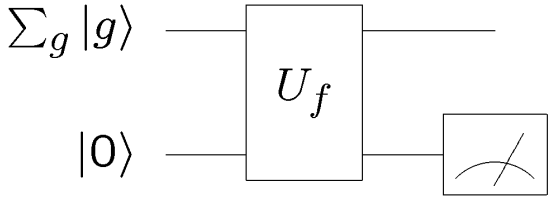
Efficient: If runs in time polylog($|\mathcal{G}|$)



Start Simple

$$\mathcal{G} = \mathbb{Z}_2 = \{e, r\}$$

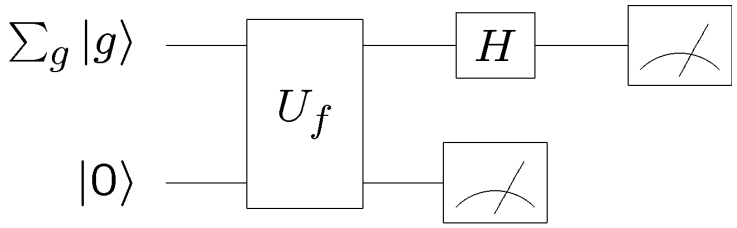
	e	r
e	e	r
r	r	e



Two possible hidden subgroups

$$\mathcal{H}_t = \{e\} \text{ or } \mathcal{H}_2 = \{e, r\}$$

	$\mathcal{H}_t$	$\mathcal{H}_2$
50%	$\|0\rangle\langle 0\|$	$\|+\rangle\langle +\|$
50%	$\|1\rangle\langle 1\|$	$\|\pm\rangle = \frac{1}{\sqrt{2}}(\|0\rangle \pm \|1\rangle)$
	$\rho_t = \frac{1}{2} I$	$\rho_2 = \|+\rangle\langle +\|$



Symmetr

I give you a symmetric probability distribution

$$T\vec{p} = \vec{p}$$

how does this help you?

I give you a symmetric quantum density matrix

$$T\rho T^\dagger = \rho$$

how does this help you?

T, ρ can be simultaneously diagonalized!

$$\rho_t = \frac{1}{2} I$$

$$\rho_2 = |+\rangle\langle +|$$

$$X\rho_h X^\dagger = \rho_h \Rightarrow$$

diagonalize in $|\pm\rangle$ basis

One More Outrageous Statements

The Symmetry Speedup Hypothesis of Quantum Computing

Every exponential quantum speedup in quantum algorithms is fundamentally the result of exploiting symmetries.



Hidden Subgroup Divide

Successes
- abelian groups
- almost abelian groups
- near Hamiltonian groups
- normal subgroups
- small commutator subgroup
- solvable groups of bounded exponent and of bounded derived series
- extraspecial groups
- groups of nilpotency class at most 2

Specific groups

$\mathbb{Z}_2^n \rtimes \mathbb{Z}_2$

$\mathbb{Z}_{p^k}^n \rtimes \mathbb{Z}_2$

some $\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$

including

the Heisenberg group

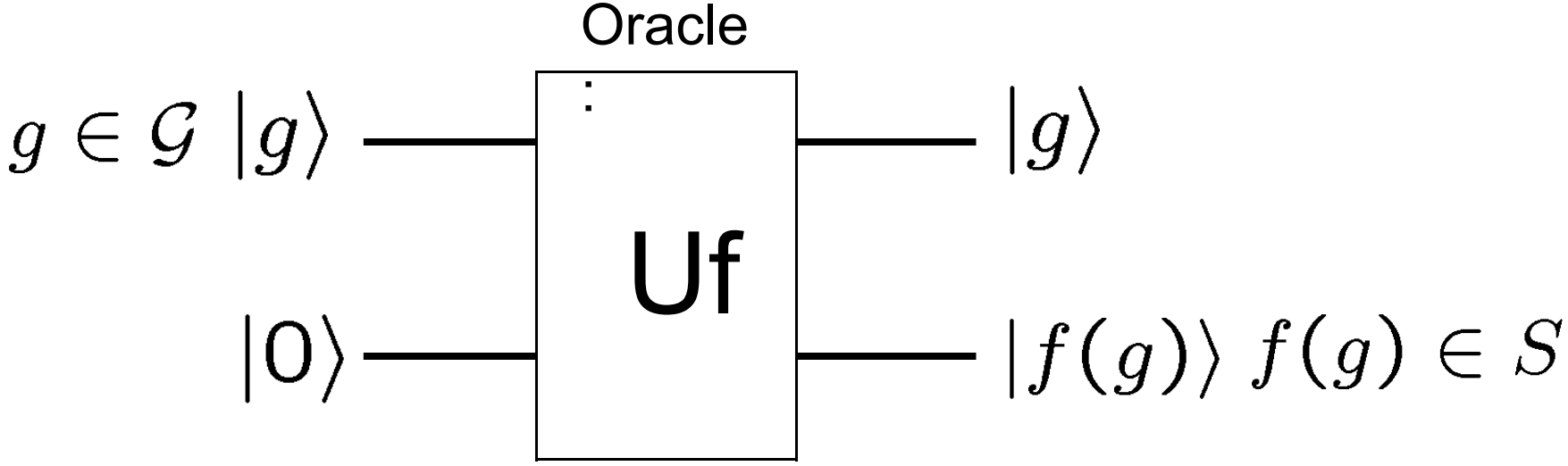
$\mathbb{Z}_N \rtimes \mathbb{Z}_p, N/p = \text{polylog} N$

Failures
- symmetric group (graph ismorphism!)
- dihedral group (shortest vector!)
- Gn, fixed G. Even for G=S3!

Pessimism
- in standard approach, multiregister measurements required
- optimal measurement for dihedral HSP leads naturally to subset sum problem
- Clebsch-Gordan sieves cannot solve symmetric group HSP

Subexponential algorithms
- Dihedral
- Gn, fixed G

Standard Query Model



Query in equal superposition over group elements

$$\frac{1}{\sqrt{|\mathcal{G}|}} \sum_{g \in \mathcal{G}} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|\mathcal{G}|}} |g\rangle|f(g)\rangle$$

Hidden subgroup state:

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}| \qquad |g\mathcal{H}\rangle := \frac{1}{\sqrt{\mathcal{H}}} \sum_{h \in \mathcal{H}} |gh\rangle$$

HSP Symmetries

Symmetry

left regular representation: $D_L(g)|g'\rangle = |gg'\rangle$

$$D_L(g)\rho_{\mathcal{H}}D_L(g^{-1}) = \rho_{\mathcal{H}}$$

Transform/Consequence

QFT over $\mathcal{G}$

$$U = \sum_{g\in\mathcal{G}}\sum_{\mu}\sum_{i,j=1}^{d_\mu}\sqrt{\frac{d_\mu}{|\mathcal{G}|}}[D_\mu(g)]_{i,j}|\mu,i,j\rangle\langle g|$$

block diagonal in QFT basis:

$$\rho_{\mathcal{H}} = \bigoplus_{\mu} I_{d_\mu}\otimes\sigma_{\mu,\mathcal{H}}$$

This symmetry is “enough” to solve the abelian HSP and case where subgroups are normal.

HSP Multiple Registers

Given *m* copies of HSP can we perform some measurement on these states to identify *H*?

$$\rho_{\mathcal{H}}^{\otimes m}$$

Hallgren, Moore, Roetteler, Russell Sen (2005): $m > O(\log |H|)$

Multi-copy (multi-register) measurements required.

Symmetry

symmetric group

$$S_\pi |i_1, \ldots, i_n\rangle = |i_{\pi(1)}, \ldots, i_{\pi(n)}\rangle$$

$$S_\pi \rho_{\mathcal{H}}^{\otimes m} S_\pi^{-1} = \rho_{\mathcal{H}}^{\otimes m}$$

Transform/Consequence

Schur Transform $\rho_{\mathcal{H}}^{\otimes m} = \bigoplus_Y I_{d_Y} \otimes \sigma_{Y,\mathcal{H}}$

Measuring *Y* not enough to solve HSP [Childs, Wocjan 2006]

Our Approach, Recast!

Two subgroups, $\mathcal{H}_1$ and $\mathcal{H}_2$, are **conjugate** if there exists an element of the group $g \in \mathcal{G}$ such that

$$\mathcal{H}_1 = \{ghg^{-1}, h \in \mathcal{H}_2\}$$

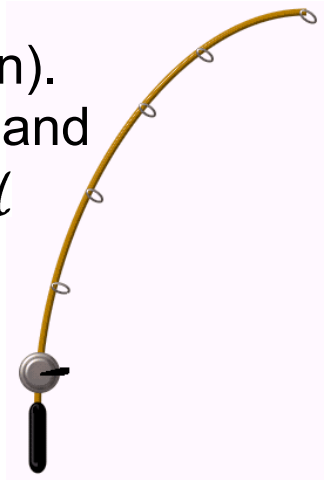


Equivalence relation among the subgroups of $\mathcal{G}$

$$\mathcal{H}_i \equiv_{\mathcal{G}} \mathcal{H}_j$$

Hidden Subgroup Conjugacy Problem

Setup: Fix a group $\mathcal{G}$ (known) and a subgroup $\mathcal{H} \subset \mathcal{G}$ (unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$. We say that f hides $\mathcal{H}$.
Problem: **identify the conjugate subgroup class** of $\mathcal{H}$.
Efficient: If runs in time polylog($|\mathcal{G}|$)



Why This Approach?

- HSCP = HSP for many “solved” cases
  Abelian, Normal subgroups, more (this talk)

- HSCP possesses more symmetry than HSP

  Symmetry in multi-register case (as REQUIREI

  Symmetry related to Clebsch-Gordan transforn

- HSCP is the relevant problem for Graph Isomorphism

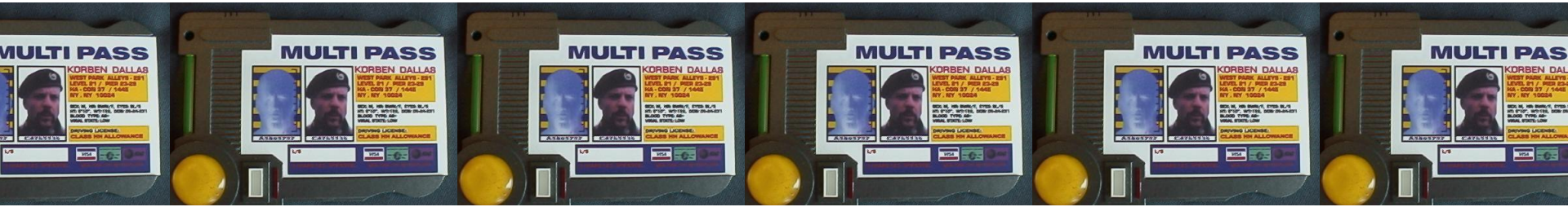


Multicopy HSP and HSCP

Algorithms should be able to distinguish:

	Single query	Multiple queries
HSP	$\rho_{\mathcal{H}}$	$\rho_{\mathcal{H}}^{\otimes m}$
HSCP	$\rho_{[\mathcal{H}]} = \frac{1}{\lvert\mathcal{G}\rvert} \sum_{g \in \mathcal{G}} \rho_{\mathcal{H}^g}$	$\rho_{[\mathcal{H}],m} = \frac{1}{\lvert\mathcal{G}\rvert} \sum_{g \in \mathcal{G}} \rho_{\mathcal{H}^g}^{\otimes m}$

$$\mathcal{H}^g = \{ghg^{-1}, h \in \mathcal{H}\}$$



HSCP Symmetries

State	Symmetry	Transform/Consequence
$\rho_{[\mathcal{H}]}$	right regular representation $D_R(g)\|g'\rangle = \|g'g^{-1}\rangle$ $D_R(g)\rho_{[\mathcal{H}]}D_R(g^{-1}) = \rho_{[\mathcal{H}]}$	QFT over $\mathcal{G}$ $\rho_{[\mathcal{H}]} = \bigoplus_{\mu} p_{\mu,\mathcal{H}} I_{d_\mu} \otimes I_{d_\mu}$
$\rho_{[\mathcal{H}],m}$	diagonal action of right regular representation	Clebsch-Gordan transform over $\mathcal{G}$

$$D_R(g)^{\otimes m}\rho_{[\mathcal{H}],m}D_R(g^{-1})^{\otimes m} = \rho_{[\mathcal{H}],m}$$

$$\rho_{[\mathcal{H}]}^{\otimes m} = \bigoplus_{\mu_1,\ldots,\mu_m} \left(I_D \otimes \bigoplus_{\mu} \left(R_{[\mathcal{H}]} \otimes I_{d_\mu} \right) \right)$$

Clebsch-Gordan Transform

Clebsch-Gordan series

$$D_\mu(g) \otimes D_\nu(g) \cong \bigoplus_\gamma n^\gamma_{\mu,\nu} D_\gamma(g) \qquad n^\gamma_{\mu,\nu} \in \mathbb{Z}^+$$

mulitplicity

If all $n^\gamma_{\mu,\nu} = 0, 1$ then we say "multiplicity-free"

Clebsch-Gordan transform

[m=2]

$$U_{CG}[D_\mu(g) \otimes D_\nu(g)]U^\dagger_{CG} = \bigoplus_\gamma I_{n^\gamma_{\mu,\nu}} \otimes D_\gamma(g)$$

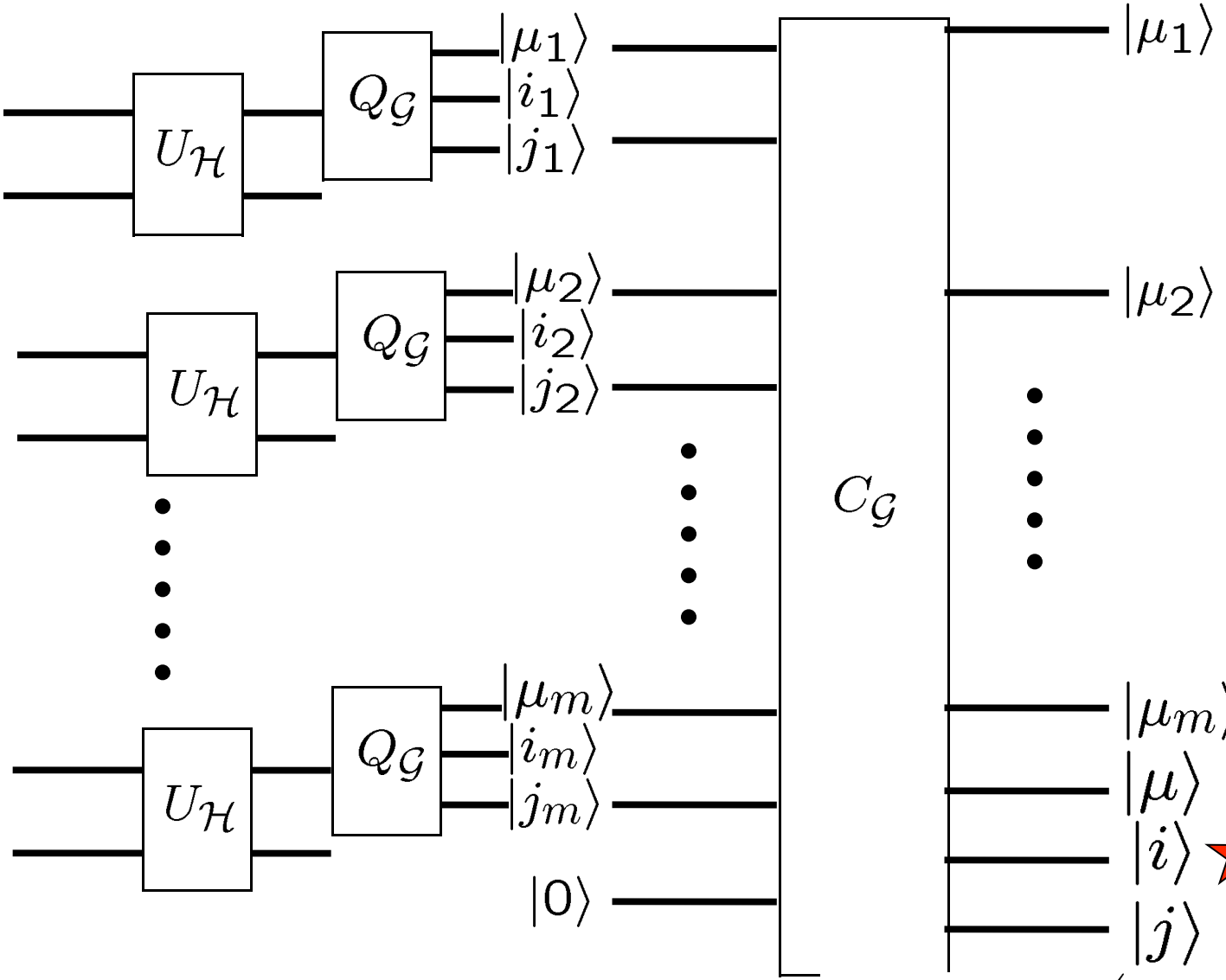
multiplicity space irrep space

[general m]

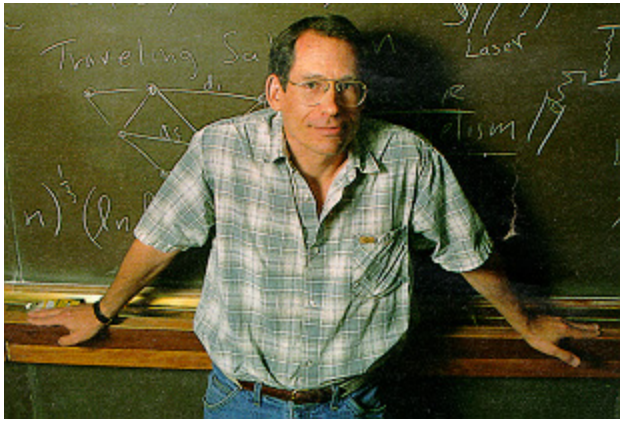
$$U_{CG}[\bigotimes_i D_{\mu_i}(g)]U^\dagger_{CG} = \bigoplus_\gamma I_{\prod_{i=1}^{m-1} n^\gamma_{\mu_i,\mu_{i+1}}} \otimes D_\gamma(g)$$

General HSCP Circuit





$$\rho_{[\mathcal{H}]}^{\otimes m} = \bigoplus_{\mu_1,\ldots,\mu_m} \left(I_D \otimes \bigoplus_{\mu} \left(R_{[\mathcal{H}]} \otimes I_{d_\mu} \right) \right)$$



…Is the CG transform really useful for HSP algorithms?

Simon's Problem

Simon: Given a function f:{0,1}^n →{0,1}^n such that f(x)=f(x⊕y), y≠0
Problem: Find y∈{0,1}^n by querying f

Simon's algorithm:

1. Query f in superposition:

$$\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \otimes |f(x)\rangle$$

2 .Discard second register:

$$\frac{1}{\sqrt{2}}(|x\rangle + |x \oplus y\rangle)$$

3. Hadamard qub

$$\frac{1}{\sqrt{2^{n-1}}} \sum_{s | s \cdot y = 0} |s\rangle \quad s \cdot y = s_1 y_1 + \cdots + s_n y_n \mod 2$$

Measuring we obtain *s* such that $s \cdot y = 0$

(With high probability) repeating this n-1 times produces n-1 linearly independent *s*'s. Using Gaussian elimination we can then obtain *y*

Simon's Problem Redux

Simon: Hidden subgroup problem over $\mathbb{Z}_2^n$ with subgroups order two

QFT on $\mathbb{Z}_2$ is nothing more than the Hadamard transform

Regular representations of $\mathbb{Z}_2$

$$D(0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad D(1) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Hadamard block-diagonalizes these to trivial and alternating irreps:

$$HD(0)H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad HD(1)H = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad HD(b)H = [1] \oplus ([-1)^b]$$

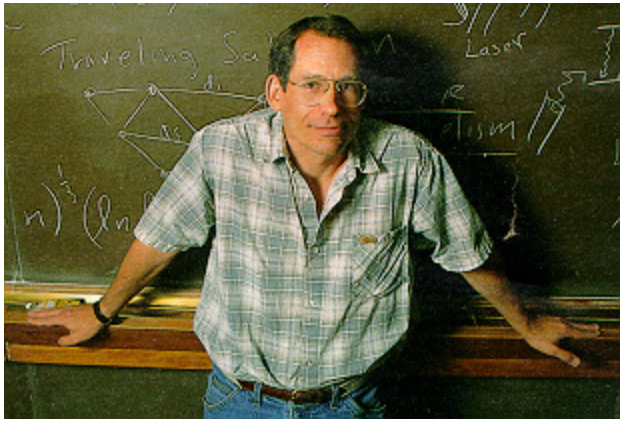
But what is Gaussian Elimination?

$$\begin{array}{r} 01001000 \\ +\ 1 \\ \hline 101001011 \\ 111011010 \end{array}$$

QFT transformed bit vectors

$$[t] \otimes [t] = [t]$$
$$[t] \otimes [a] = [a]$$
$$[a] \otimes [t] = [a]$$
$$[a] \otimes [a] = [t]$$

Gaussian elimination is nothing more than CG transforms over $\mathbb{Z}_2^n$



…Is the CG transform really useful for non-abelian HSP algorithms?

Heisenberg Group

Semidirect product group: $\mathbb{Z}_p^2 \rtimes \mathbb{Z}_p$

Group of upper triangular matrices:

$$\begin{bmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{bmatrix} \quad x, y, z \in \mathbb{Z}_p$$

(Related to theory of discrete radar)

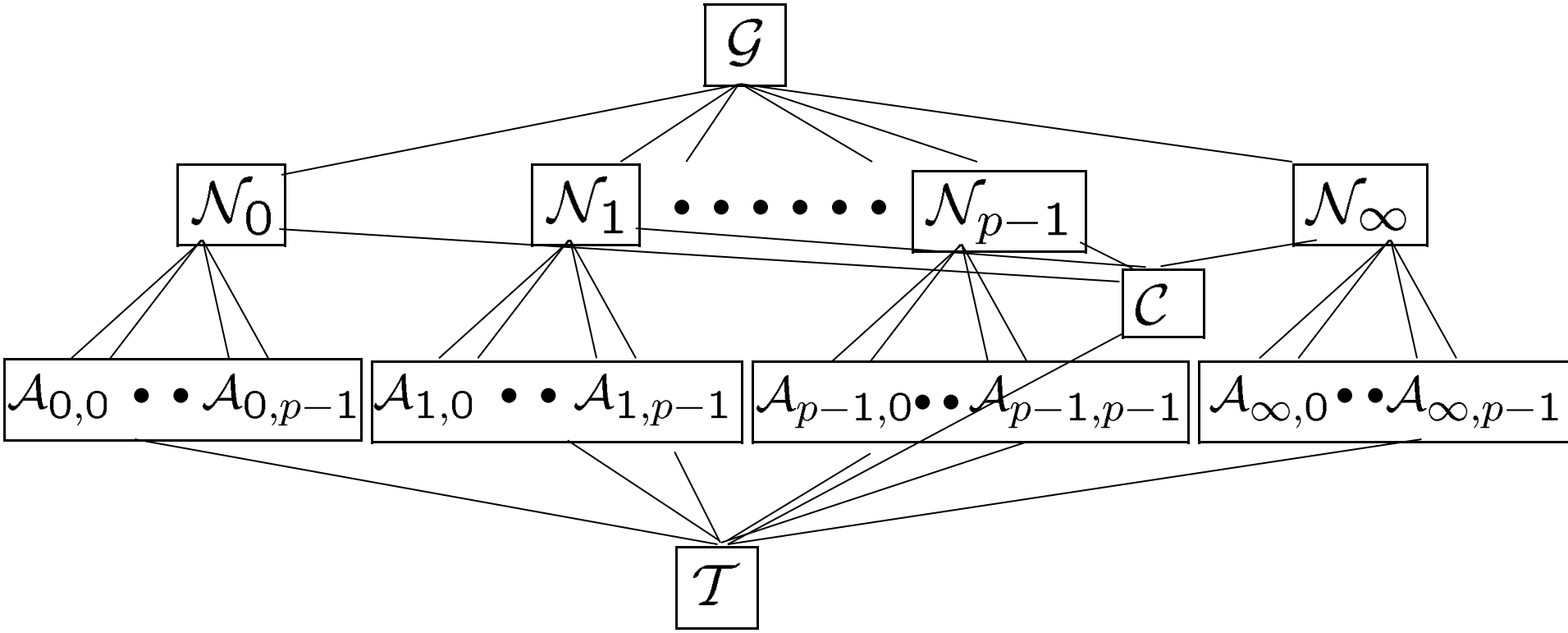


Bohr and Heisenberg

Efficient quantum algorithm for HSP over Heisenberg group
[Bacon, Childs, van Dam quant-ph/0504083]

HSP and HSCP

Heisenberg group



$(\mathcal{N}_\alpha, \mathcal{C}, \mathcal{G}, \mathcal{T}) \triangleleft \mathcal{G}$ $\quad \mathcal{A}_{\alpha,i} \not\triangleleft \mathcal{G}$ $\quad \mathcal{A}_{\alpha,i} \triangleleft \mathcal{N}_\alpha$

Solving HSCP $\Rightarrow$ Identifying α *of* $\mathcal{A}_{\alpha,i}$ normal HS algorithm

Result: Heisenberg HSP = Heisenberg HSCP

[Bacon, quant-ph/0612107, submitted]

Heisenberg CG

Representation Theory

p2 1D irreps $D_{a,b}(g), a, b \in \mathbb{Z}_p$

*p (p-1)*D irreps $D_c(g), c \in \mathbb{Z}^*$

Clebsch-Gordan series:

$$D_{a_1,b_1} \otimes D_{a_2,b_2} = D_{a_1+a_2,b_1+b_2}$$

$$D_{a_1,b_1} \otimes D_{c_2} = D_{c_2}$$

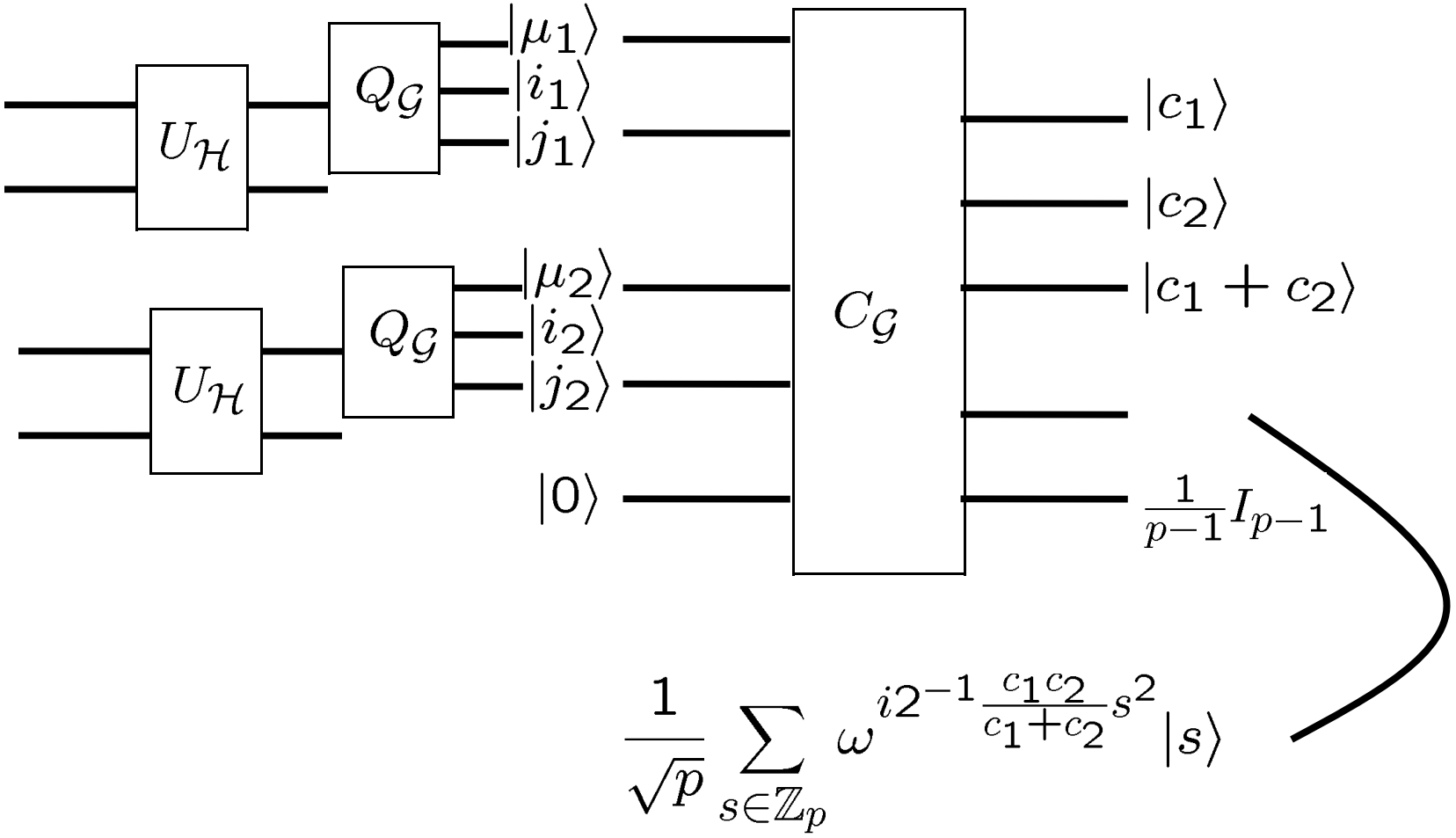
$$D_{c_1} \otimes D_{c_2} = pD_{c_1+c_2} \qquad c_1 + c_2 \neq 0 \mod p$$

$$D_{c_1} \otimes D_{-c_1} = \oplus_{a,b \in \mathbb{Z}_p} D_{a,b}$$

Nontrivial multiplicity space for (*p-1*)D irreps!

Heisenberg HSCP, CG Way

[Bacon, arXiv:quant-ph/0612107]



i labels set of conjugate subgroups

Heisenberg HSCP, CG Way

$$\frac{1}{\sqrt{p}} \sum_{s \in \mathbb{Z}_p} \omega^{i 2^{-1} \frac{k_1 k_2}{k_1 + k_2} s^2} |s\rangle$$

i can be found by performing

$$|t^2\rangle \rightarrow \frac{1}{\sqrt{2}}(|t\rangle + |-t\rangle)$$

and performing a
QFT

Clebsch-Gordan transform over Heisenberg group is related to “symmetry” which allows for efficient quantum algorithm for the Heisenberg HSP.

Heisenberg HSP Comparison

PGM/optimal measurement approach [Bacon,Childs, van Dam 05]

$$\rho_{\mathcal{H}} \qquad Pr(succ) = O\left(\frac{1}{p}\right)$$

$$\rho_{\mathcal{H}} \otimes \rho_{\mathcal{H}} \qquad Pr(succ) = O(1)$$

Two copy state:

$$\frac{1}{p} \sum_{b_1, b_2 \in \mathbb{Z}_p} \omega_p^{(b_1 x_1 + b_2 x_2) d_1 + \left(\frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2\right) d_2} |b_1, b_2\rangle$$

CG symmetry reduces this to

$$\frac{1}{\sqrt{p}} \sum_{s \in \mathbb{Z}_p} \omega^{i 2^{-1} \frac{k_1 k_2}{k_1 + k_2} s^2} |s\rangle$$

Nonabelian Simon's Problem

Generalize Simon's problem to $\mathcal{G}^n$ where $\mathcal{G}$ is a fixed size nonabelian group with an order two subgroup.

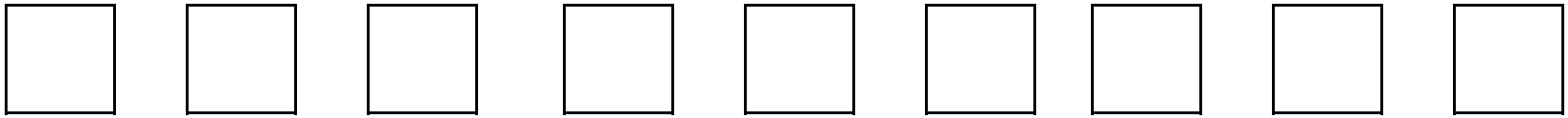
Alagic, Moore, Russell (2006) found a subexponential, but still superpolynomial time algorithm for this problem.

Their algorithm relies on weak-Clebsch-Gordan sampling

What about information in the multiplicity register?

A New Quantum HSP Algorithm

Simon's problem over the n-fold product of the dihedral group of order eight

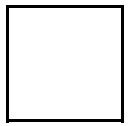


Dihedral Group of Order Eight

Group of symmetries of the square:

Rotation ≡ s, s4=e

Reflection ≡ r, r2=e and rsr=s-1



Element of D4: rtsk, where t∈Z2,s∈Z4

Consider the four two dimensional representations (reducible):

$$D_j(r^t s^k) = \begin{bmatrix} \omega^{jk}\delta_{t,0} & \omega^{-jk}\delta_{t,1} \\ \omega^{jk}\delta_{t,1} & \omega^{-jk}\delta_{t,0} \end{bmatrix}$$

Clebsch-Gordan series

$$D_{j_1}(g) \otimes D_{j_2}(g) = D_{j_1+j_2}(g) \oplus D_{j_1-j_2}(g)$$

A Warm-up

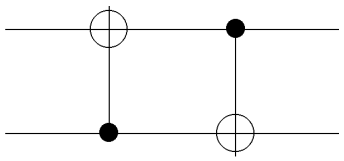
HSCP on D4, distinguish parity of k,
{e,rsk}
Partial QFT + measurement yields random $\mu \in_U \mathbb{Z}_4$

$$|\psi_{\mu,l}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i^{\mu l}|1\rangle)$$

Note that if $\mu = 2$ a Hadamard will reveal parity of *l*
if $\mu = 0$ information about *l* exists

Two copies, $\mu_1, \mu_2 \in \{1, 3\}$

$$\frac{1}{2}(|0\rangle + i^{\mu_1 l}|1\rangle) \otimes (|0\rangle + i^{\mu_2 l}|1\rangle)$$



CNOT+SWAP:

$$|0\rangle \otimes (|0\rangle + i^{(\mu_1+\mu_2)l}|1\rangle) + i^{\mu_2 l}|1\rangle \otimes (|0\rangle + i^{(\mu_1-\mu_2)l}|1\rangle)$$

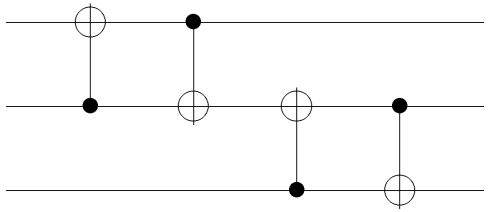
Measure first register, sometimes get $\mu_{tot} = 2$
This is just a Clebsch-Gordan sieve, nothing really new here

A Warm-up, Part II

Three copies, $\mu_1, \mu_2, \mu_3 \in \{1, 3\}$

$$\frac{1}{2\sqrt{2}}(|0\rangle + i^{\mu_1 l}|1\rangle) \otimes (|0\rangle + i^{\mu_2 l}|1\rangle) \otimes (|0\rangle + i^{\mu_3 l}|1\rangle)$$

CNOT+SWAP, cascaded



$$|00\rangle \otimes (|0\rangle + i^{(\mu_1+\mu_2+\mu_3)l}|1\rangle) + i^{\mu_3 l}|01\rangle \otimes (|0\rangle + i^{(\mu_1+\mu_2-\mu_3)l}|1\rangle)$$
$$+i^{\mu_2 l}|10\rangle \otimes (|0\rangle + i^{(\mu_1-\mu_2+\mu_3)l}|1\rangle) + i^{(\mu_2+\mu_3)l}|11\rangle \otimes (|0\rangle + i^{(\mu_1-\mu_2-\mu_3)l}|1\rangle)$$

Compute Clebsch-Gordan total and subtract off μ_1

$$(|00\rangle + i^{(\mu_2+\mu_3)l}|11\rangle) \otimes (|0\rangle + i^{(\mu_1+\mu_2+\mu_3)l}|1\rangle)|\mu_2 + \mu_3\rangle$$
$$i^{\mu_3 l}(|01\rangle + i^{(\mu_2-\mu_3)l}|10\rangle) \otimes (|0\rangle + i^{(\mu_1+\mu_2-\mu_3)l}|1\rangle)|\mu_2 - \mu_3\rangle$$

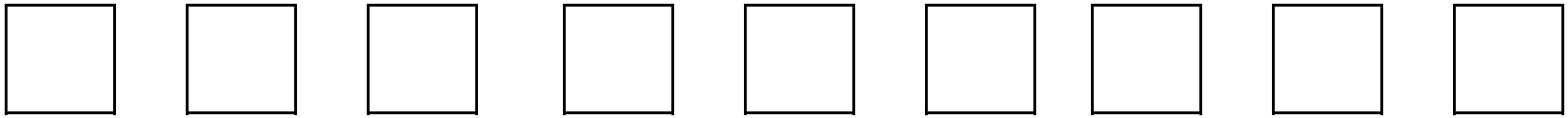
Now there is a multiplicity. Example: $\mu_1 = 1, \mu_2 = 3, \mu_3 = 1$

$$(|00\rangle + |11\rangle) \otimes (|0\rangle + i^{l}|1\rangle)|0\rangle + i(|01\rangle + (-1)^{l}|10\rangle) \otimes (|0\rangle + i^{3l}|1\rangle)|2\rangle$$

Irrep space has no information, but half the time, multiplicity space has info about *l*

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Multiplicity space algorithm for $\mathcal{D}_4^n$ to find subgroups:

$$\{e, \times_{i=1}^{n} r^{t_i} s^{l_i}\} \quad (t_i, l_i) \in \{(0,0),(1,0),(1,1),(1,2),(1,3)\}$$

Outline of algorithm:

- Determine t
- Determine parity of l
- Determine l (Given parity of l, determining l is normal HSP)

Sketch of Algorithm

Random $\mu \in_U \mathbb{Z}_4^n$ along with state $|b\rangle + i^{\mu \cdot l}|b+t\rangle$

Run this *m*=*n*+2 times $\mu_i \in_U \mathbb{Z}_4^n$

u_i parity strings μ_i of

Solve

$$\sum_{i=2}^{m} b_i u_i = 0^n \quad b_i \in \mathbb{Z}_2$$

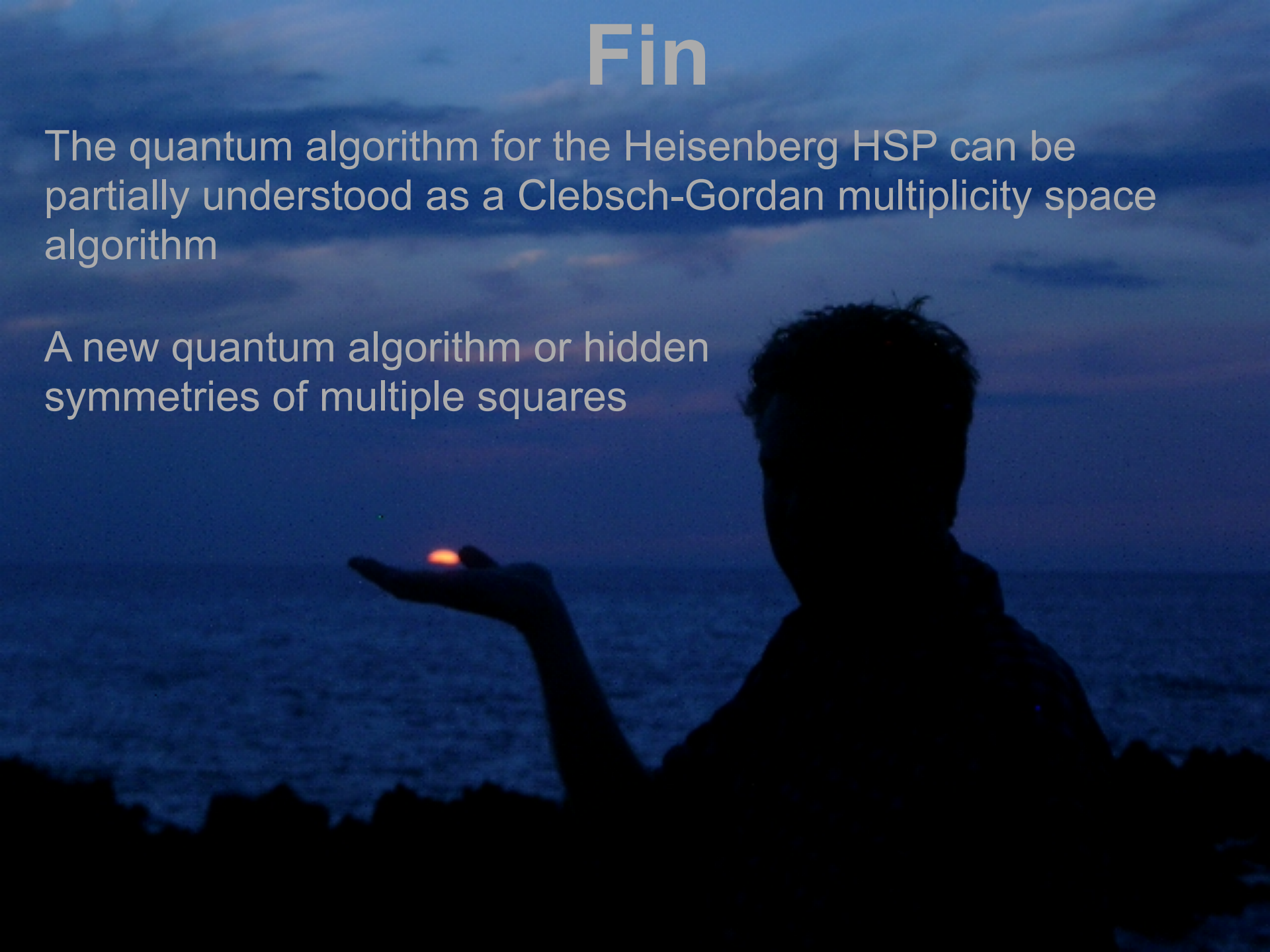
Over copies where $b_i = 1$ we can obtain a $\mu_{eff} \in \{0,2\}^n$

Repeating this multiple times we can similarly distill $\mu_{eff} = 0^n$

Then using Simon's algorithm we can determine locals where t=1.

After determining where ti=1, we can query over these bits and $\mu_{eff} \in \{0,2\}^n$

to determine parity of l



Fin

The quantum algorithm for the Heisenberg HSP can be partially understood as a Clebsch-Gordan multiplicity space algorithm

A new quantum algorithm or hidden symmetries of multiple squares